

TBI-GEU Summer Internship Program 2026

Week 1 — Topic Brief | AI-Enabled Business Intelligence

Herbal & Aromatics Sector | Intern ID: 26101136

Project Title:	AI-Enhanced Extraction Yield Intelligence System for Essential Oil Distillers
Intern ID:	26101136
Domain:	AI-Enabled Business Intelligence
Sector:	Herbal & Aromatics
Business Profile:	Alaknanda Herbal & Essential Oil Distillers, Chamoli District, Uttarakhand

Business Problem

Small-scale essential oil distillers in Uttarakhand operate with zero data visibility into batch-level extraction yields. Output varies significantly across seasons, plant varieties, altitudes, and distillation durations, yet no structured records exist to diagnose why. This prevents producers from forecasting supply, committing to buyer orders, or identifying which conditions produce the highest-quality yield.

Target Audience

Production managers and owners of small-scale herbal distillation units, and agricultural extension officers supporting Himalayan herbal producers under schemes such as AYUSH and Uttarakhand Herb Board initiatives.

Dataset(s) Identified

Primary: Custom simulated batch distillation log (600+ rows) — generated using realistic Himalayan distillation parameters. Columns: Batch_ID, Plant_Variety, Harvest_Season, Altitude_Zone, Input_Kg, Distillation_Duration_hrs, Temperature_C, Output_ml, Yield_Efficiency_pct, Quality_Grade.

Secondary: Oil Seeds Production & Yield Analysis — Kaggle (kaggle.com/datasets/ranamuhhammadusman/oil-seeds-dataset-production-and-yield-analysis). ~800 rows. Key columns: crop_name, state, year, area_hectares, production_tonnes, yield_kg_per_hectare.

Key Business Questions

1. Which plant variety and harvest season combination produces the highest extraction yield efficiency?
2. How does altitude zone and distillation duration interact to affect output volume and quality grade?
3. Can an AI model predict expected yield (ml) from input parameters before a batch begins distillation?

Proposed Tool Stack

Python (pandas, numpy) — data cleaning and transformation | matplotlib / seaborn — exploratory and correlation visualisation | scikit-learn — yield prediction model (linear regression / random forest) | Looker Studio — interactive dashboard for production managers | Claude / Gemini AI — plain-language insight summaries from batch data.

Expected Capstone Output

A fully interactive BI dashboard showing batch yield trends, seasonal patterns, and key efficiency drivers, paired with an AI-generated insight report that a non-technical production manager can act on directly. The final recommendation will identify the top 3 controllable variables that most improve yield efficiency and suggest a standard operating protocol for distillation batches.